R Programming Practical Examination

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Part A: Basic Operations

```
## A1

v <- seq(5, 50, by=5)
length(v)
```

```
## [1] 10
```

```
sum(v)
```

```
## [1] 275
```

```
mean(v)
```

```
## [1] 27.5
```

```
## A2

w <- rep(c(1, 2, 3), times = 4)
z <- rep(c(1, 2, 3), each = 4)
w
```

```
##  [1] 1 2 3 1 2 3 1 2 3 1 2 3
```

```
z
```

```
##  [1] 1 1 1 1 2 2 2 2 3 3 3 3
```

```
## A3

17 %% 5
```

```
## [1] 2
```

```
17 %/% 5
```

```
## [1] 3
```

```
2^10
```

```
## [1] 1024
```

```
sqrt(144)
```

```
## [1] 12
```

```
round(pi, 3)
```

```
## [1] 3.142
```

```
## A4

a <- c(2, 4, 6, 8)
b <- c(1, 3, 5, 7)

a + b
```

```
## [1]  3  7 11 15
```

```
a * b
```

```
## [1]  2 12 30 56
```

```
a > b
```

```
## [1] TRUE TRUE TRUE TRUE
```

```
sum(a * b)
```

```
## [1] 100
```

```
## A5

paste("Student", 1:3, sep = "-")
```

```
## [1] "Student-1" "Student-2" "Student-3"
```

```
typeof(42)
```

```
## [1] "double"
```

```
typeof(42L)
```

```
## [1] "integer"
```

```
typeof("42")
```

```
## [1] "character"
```

```
typeof(41i)
```

```
## [1] "complex"
```

```
typeof(TRUE)
```

```
## [1] "logical"
```

Part B: Data Frames and Conditional Statements

```
## B1

students <- data.frame(
id = 1:8,
name = c("Alice", "Bruno", "Chen", "Dana", "Elif", "Farid", "Grace", "Hiro"),
sex = c("F", "M", "M", "F", "F", "M", "F", "M"),
section = c("A", "B", "A", "C", "B", "C", "A", "B"),
midterm = c(78, 85, 92, 67, 74, 88, 95, 59),
final = c(82, 79, 96, 71, 80, 91, 93, 65),
attendance = c(95, 88, 100, 72, 90, 85, 98, 60),
stringsAsFactors = FALSE
)
```

```
## B2

str(students)
```

```
## 'data.frame':    8 obs. of  7 variables:
##  $ id        : int  1 2 3 4 5 6 7 8
##  $ name      : chr  "Alice" "Bruno" "Chen" "Dana" ...
##  $ sex       : chr  "F" "M" "M" "F" ...
##  $ section   : chr  "A" "B" "A" "C" ...
##  $ midterm   : num  78 85 92 67 74 88 95 59
##  $ final     : num  82 79 96 71 80 91 93 65
##  $ attendance: num  95 88 100 72 90 85 98 60
```

```
dim(students)
```

```
## [1] 8 7
```

```
colnames(students)
```

```
## [1] "id"         "name"       "sex"        "section"    "midterm"
## [6] "final"      "attendance"
```

```
summary(students)
```

```
##        id              name               sex             section     midterm
##  Min.   :1.00   Length   :8   Length   :8   Length   :8   Min.   :59.00
##  1st Qu.:2.75   N.unique :8   N.unique :2   N.unique :3   1st Qu.:72.25
##  Median :4.50   N.blank  :0   N.blank  :0   N.blank  :0   Median :81.50
##  Mean   :4.50   Min.nchar:4   Min.nchar:1   Min.nchar:1   Mean   :79.75
##  3rd Qu.:6.25   Max.nchar:5   Max.nchar:1   Max.nchar:1   3rd Qu.:89.00
##  Max.   :8.00                                           Max.   :95.00
##      final         attendance
##  Min.   :65.00   Min.   : 60.00
##  1st Qu.:77.00   1st Qu.: 81.75
##  Median :81.00   Median : 89.00
##  Mean   :82.12   Mean   : 86.00
##  3rd Qu.:91.50   3rd Qu.: 95.75
##  Max.   :96.00   Max.   :100.00
```

```
## B3

students$average <- round((students$midterm + students$final) / 2, 1)
students
```

```
##   id  name sex section midterm final attendance average
## 1  1 Alice   F       A      78    82         95    80.0
## 2  2 Bruno   M       B      85    79         88    82.0
## 3  3  Chen   M       A      92    96        100    94.0
## 4  4  Dana   F       C      67    71         72    69.0
## 5  5  Elif   F       B      74    80         90    77.0
## 6  6 Farid   M       C      88    91         85    89.5
## 7  7 Grace   F       A      95    93         98    94.0
## 8  8  Hiro   M       B      59    65         60    62.0
```

```
## B4

students$passed <- students$average >= 70 & students$attendance >= 75
students
```

```
##   id  name sex section midterm final attendance average passed
## 1  1 Alice   F       A      78    82         95    80.0   TRUE
## 2  2 Bruno   M       B      85    79         88    82.0   TRUE
## 3  3  Chen   M       A      92    96        100    94.0   TRUE
## 4  4  Dana   F       C      67    71         72    69.0  FALSE
## 5  5  Elif   F       B      74    80         90    77.0   TRUE
## 6  6 Farid   M       C      88    91         85    89.5   TRUE
## 7  7 Grace   F       A      95    93         98    94.0   TRUE
## 8  8  Hiro   M       B      59    65         60    62.0  FALSE
```

```
## B5

students$standing <- ifelse(students$average >= 90, "Honors", ifelse(students$average >= 75, "Good", "N
students
```

```
##   id  name sex section midterm final attendance average passed      standing
## 1  1 Alice   F       A      78    82         95    80.0   TRUE          Good
## 2  2 Bruno   M       B      85    79         88    82.0   TRUE          Good
## 3  3  Chen   M       A      92    96        100    94.0   TRUE        Honors
## 4  4  Dana   F       C      67    71         72    69.0  FALSE Needs Support
## 5  5  Elif   F       B      74    80         90    77.0   TRUE          Good
## 6  6 Farid   M       C      88    91         85    89.5   TRUE          Good
## 7  7 Grace   F       A      95    93         98    94.0   TRUE        Honors
## 8  8  Hiro   M       B      59    65         60    62.0  FALSE Needs Support
```

```
## B6

classify <- function(score) {
if (score >= 85) "Distinction"
else if (score >= 70) "Merit"
else if (score >= 50) "Pass"
else "Fail"
}
sapply(c(90, 73, 55, 40), classify)
```

```
## [1] "Distinction" "Merit"       "Pass"        "Fail"
```

Part C: rbind() and cbind()

```
## C1

new_students <- data.frame(
  id = c(9, 10),
  name = c("Ines", "Jamal"),
  sex = c("F", "M"),
  section = c("C", "A"),
  midterm = c(81, 70),
  final = c(84, 68),
  attendance = c(92, 80),
  stringsAsFactors = FALSE
)

new_students$average <- round((new_students$midterm + new_students$final) / 2, 1)
new_students$passed <- new_students$average >= 70 & new_students$attendance >= 75
new_students$standing <- ifelse(new_students$average >= 90, "Honors", ifelse(new_students$average >= 75
```

```r
students_all <- rbind(students, new_students)
students_all
```

```
##    id  name sex section midterm final attendance average passed      standing
## 1   1 Alice   F       A      78    82         95    80.0   TRUE          Good
## 2   2 Bruno   M       B      85    79         88    82.0   TRUE          Good
## 3   3  Chen   M       A      92    96        100    94.0   TRUE        Honors
## 4   4  Dana   F       C      67    71         72    69.0  FALSE Needs Support
## 5   5  Elif   F       B      74    80         90    77.0   TRUE          Good
## 6   6 Farid   M       C      88    91         85    89.5   TRUE          Good
## 7   7 Grace   F       A      95    93         98    94.0   TRUE        Honors
## 8   8  Hiro   M       B      59    65         60    62.0  FALSE Needs Support
## 9   9  Ines   F       C      81    84         92    82.5   TRUE          Good
## 10 10 Jamal   M       A      70    68         80    69.0  FALSE Needs Support
```

```r
## C2

quiz <- c(8, 7, 10, 6, 7, 9, 10, 5, 8, 6)
students_all <- cbind(students_all, quiz = quiz)
class(students_all)
```

```
## [1] "data.frame"
```

```r
## C3

m <- cbind(midterm = students_all$midterm, final = students_all$final)
m <- rbind(m, mean = colMeans(m))
m
```

```
##      midterm final
##         78.0  82.0
##         85.0  79.0
##         92.0  96.0
##         67.0  71.0
##         74.0  80.0
##         88.0  91.0
##         95.0  93.0
##         59.0  65.0
##         81.0  84.0
##         70.0  68.0
## mean    78.9  80.9
```

```r
## C4

bad <- data.frame(x = 1, y = 2)

tryCatch(
  rbind(students, bad),
  error = function(e) {
    message("rbind failed: ", conditionMessage(e))
    NULL
  }
)
```

```
## rbind failed: numbers of columns of arguments do not match
```

```
## NULL
```

```
## rbind() throws an error when column names don't match because it doesn't know how to align the colum
```

Part D: Subsetting

```
## D1

students_all$name
```

```
##  [1] "Alice" "Bruno" "Chen"  "Dana"  "Elif"  "Farid" "Grace" "Hiro"  "Ines"
## [10] "Jamal"
```

```
students_all[["name"]]
```

```
##  [1] "Alice" "Bruno" "Chen"  "Dana"  "Elif"  "Farid" "Grace" "Hiro"  "Ines"
## [10] "Jamal"
```

```
students_all[, "name"]
```

```
##  [1] "Alice" "Bruno" "Chen"  "Dana"  "Elif"  "Farid" "Grace" "Hiro"  "Ines"
## [10] "Jamal"
```

```
## D2

students_all[2:5, c("name", "midterm", "final")]
```

```
##    name midterm final
## 2 Bruno      85    79
## 3  Chen      92    96
## 4  Dana      67    71
## 5  Elif      74    80
```

```
## D3

students_all[students_all$attendance > 90, ]
```

```
##   id  name sex section midterm final attendance average passed standing quiz
## 1  1 Alice   F       A      78    82         95    80.0   TRUE     Good    8
## 3  3  Chen   M       A      92    96        100    94.0   TRUE   Honors   10
## 7  7 Grace   F       A      95    93         98    94.0   TRUE   Honors   10
## 9  9  Ines   F       C      81    84         92    82.5   TRUE     Good    8
```

```
## D4

subset(students_all, sex == "F" & (section == "A" | section == "C") & final >= 75, select = c("name", "
```

```
##    name section final
## 1 Alice       A    82
## 7 Grace       A    93
## 9  Ines       C    84
```

```
## D5

students_all$name[which.max(students_all$average)]
```

```
## [1] "Chen"
```

```
students_all$name[which.min(students_all$average)]
```

```
## [1] "Hiro"
```

```
## D6

sorted_students <- students_all[order(-students_all$average), ]
head(sorted_students, 3)
```

```
##   id  name sex section midterm final attendance average passed standing quiz
## 3  3  Chen   M       A      92    96        100    94.0   TRUE   Honors   10
## 7  7 Grace   F       A      95    93         98    94.0   TRUE   Honors   10
## 6  6 Farid   M       C      88    91         85    89.5   TRUE     Good    9
```

Part E: Basic Factors

```
## E1

students_all$section <- factor(students_all$section)
students_all$sex <- factor(students_all$sex)

levels(students_all$section)
```

```
## [1] "A" "B" "C"
```

```
levels(students_all$sex)
```

```
## [1] "F" "M"
```

```
## E2

table(students_all$sex)
```

```
##
## F M
## 5 5
```

```
table(students_all$section)
```

```
##
## A B C
## 4 3 3
```

```
table(students_all$sex, students_all$section)
```

```
##
##     A B C
##   F 2 1 2
##   M 2 2 1
```

```
## E3

as.integer(students_all$sex)
```

```
##  [1] 1 2 2 1 1 2 1 2 1 2
```

```
## The numbers just represent the factor levels in alphabetical order (1 for "F", 2 for "M").
```

```
## E4

students_all$sex_label <- factor(students_all$sex,
                                 levels = c("F", "M"),
```

```
                                  labels = c("Female", "Male"))
levels(students_all$sex_label)
```

```
## [1] "Female" "Male"
```

```
## E5

summary(students_all$sex_label)
```

```
## Female   Male
##      5      5
```

```
mean(students_all$average[students_all$sex_label == "Female"])
```

```
## [1] 80.5
```

```
mean(students_all$average[students_all$sex_label == "Male"])
```

```
## [1] 79.3
```